

ToughTorq Upper-Locking Hydraulic Nuts - 1

TTNM Series | Ultra-high-pressure locking for critical bolted joints

Model	Bolt	Pressure	Max Load	OD	Height	Stroke	Temp
TTNM-M33	M33	21,756 psi	48,559 lbf	2.91 in	2.52 in	0.24 in	100 C
TTNM-M36	M36	21,756 psi	55,303 lbf	3.15 in	2.8 in	0.16 in	100 C
TTNM-M39	M39	21,756 psi	66,543 lbf	3.43 in	2.95 in	0.16 in	100 C
TTNM-M42	M42	21,756 psi	76,435 lbf	3.7 in	3.19 in	0.24 in	100 C
TTNM-M45	M45	21,756 psi	91,272 lbf	3.98 in	3.46 in	0.24 in	100 C
TTNM-M48	M48	21,756 psi	102,738 lbf	4.21 in	3.5 in	0.31 in	100 C
TTNM-M52	M52	21,756 psi	124,319 lbf	4.61 in	3.78 in	0.31 in	100 C
TTNM-M56	M56	21,756 psi	147,475 lbf	5.04 in	3.94 in	0.31 in	100 C
TTNM-M64	M64	21,756 psi	183,669 lbf	5.67 in	4.45 in	0.39 in	100 C
TTNM-M68	M68	21,756 psi	226,607 lbf	6.18 in	4.69 in	0.39 in	100 C

ToughTorq Upper-Locking Hydraulic Nuts - 2

TTNM Series | Ultra-high-pressure locking for critical bolted joints

Model	Bolt	Pressure	Max Load	OD	Height	Stroke	Temp
TTNM-M72	M72	21,756 psi	236,274 lbf	6.42 in	4.84 in	0.47 in	100 C
TTNM-M80	M80	21,756 psi	327,996 lbf	7.36 in	5.59 in	0.47 in	100 C
TTNM-M90	M90	21,756 psi	388,245 lbf	8.15 in	6.22 in	0.47 in	100 C
TTNM-M95	M95	21,756 psi	441,750 lbf	8.62 in	6.54 in	0.47 in	100 C
TTNM-M100	M100	21,756 psi	501,998 lbf	9.21 in	6.77 in	0.59 in	100 C
TTNM-M110	M110	21,756 psi	604,286 lbf	10.04 in	7.24 in	0.59 in	100 C
TTNM-M125	M125	21,756 psi	817,405 lbf	11.65 in	8.27 in	0.59 in	100 C
TTNM-M140	M140	21,756 psi	982,640 lbf	12.91 in	9.37 in	0.59 in	100 C
TTNM-M150	M150	21,756 psi	1,184,294 lbf	14.02 in	10.04 in	0.71 in	100 C

ToughTorq Bearing-Assembly Hydraulic Nuts - 1

TTHMV Series | Bearing, flywheel, propeller, and interference-fit installation/removal

Model	Thread	Pressure psi	Eff.Area	D	d1	H	Stroke	Weight
TTHMV-10E	M50x1.5	10,153-21,756	5.23 in2	4.49 in	1.99 in	1.5 in	0.2 in	6 lb
TTHMV-12E	M60x2	10,153-21,756	6.57 in2	4.92 in	2.38 in	1.5 in	0.2 in	6.2 lb
TTHMV-14E	M70x2	10,153-21,756	7.3 in2	5.31 in	2.78 in	1.5 in	0.2 in	7.1 lb
TTHMV-16E	M80x2	10,153-21,756	8.03 in2	5.75 in	3.17 in	1.5 in	0.2 in	8.2 lb
TTHMV-18E	M90x2	10,153-21,756	8.76 in2	6.14 in	3.56 in	1.5 in	0.2 in	8.8 lb
TTHMV-20E	M100x2	10,153-21,756	9.28 in2	6.54 in	3.96 in	1.5 in	0.2 in	9.7 lb
TTHMV-22E	M110x2	10,153-21,756	9.74 in2	7.01 in	4.35 in	1.5 in	0.2 in	10.9 lb
TTHMV-24E	M120x2	10,153-21,756	13.28 in2	7.4 in	4.74 in	1.5 in	0.2 in	11.6 lb
TTHMV-26E	M130x2	10,153-21,756	11.11 in2	7.8 in	5.14 in	1.5 in	0.2 in	12.5 lb
TTHMV-28E	M140x2	10,153-21,756	11.79 in2	8.19 in	5.53 in	1.54 in	0.2 in	13.2 lb
TTHMV-30E	M150x2	10,153-21,756	11.99 in2	8.66 in	5.93 in	1.54 in	0.24 in	14.6 lb

ToughTorq Bearing-Assembly Hydraulic Nuts - 2

TTHMV Series | Bearing, flywheel, propeller, and interference-fit installation/removal

Model	Thread	Pressure psi	Eff.Area	D	d1	H	Stroke	Weight
TTHMV-32E	M160x3	10,153-21,756	12.4 in2	9.13 in	6.32 in	1.57 in	0.24 in	16.8 lb
TTHMV-34E	M170x3	10,153-21,756	13.44 in2	9.61 in	6.71 in	1.61 in	0.24 in	18.5 lb
TTHMV-36E	M180x3	10,153-21,756	15.97 in2	10.08 in	7.11 in	1.61 in	0.24 in	20.2 lb
TTHMV-38E	M190x3	10,153-21,756	14.36 in2	10.63 in	7.52 in	1.65 in	0.31 in	23.1 lb
TTHMV-40E	M200x3	10,153-21,756	13.35 in2	11.1 in	7.91 in	1.69 in	0.31 in	25.4 lb
TTHMV-42E	Tr210x4	10,153-21,756	18.18 in2	11.57 in	8.35 in	1.73 in	0.35 in	27.6 lb
TTHMV-44E	Tr220x4	10,153-21,756	18.31 in2	12.05 in	8.74 in	1.73 in	0.35 in	29.8 lb
TTHMV-46E	Tr230x4	10,153-21,756	19.02 in2	12.52 in	9.13 in	1.77 in	0.35 in	33.1 lb
TTHMV-50E	Tr250x4	10,153-21,756	20.64 in2	13.46 in	9.92 in	1.81 in	0.39 in	37.5 lb
TTHMV-56E	Tr280x4	10,153-21,756	22.76 in2	14.96 in	11.1 in	1.93 in	0.47 in	48.5 lb
TTHMV-60E	Tr300x4	10,153-21,756	24.24 in2	15.91 in	11.89 in	2.01 in	0.51 in	55.1 lb

ToughTorq Bearing-Assembly Hydraulic Nuts - 3

TTHMV Series | Bearing, flywheel, propeller, and interference-fit installation/removal

Model	Thread	Pressure psi	Eff.Area	D	d1	H	Stroke	Weight
TTHMV-68E	Tr340x5	10,153-21,756	44.02 in2	17.72 in	13.46 in	2.13 in	0.55 in	69.4 lb
TTHMV-76E	Tr380x5	10,153-21,756	51.93 in2	19.61 in	15.04 in	2.28 in	0.63 in	89.3 lb
TTHMV-80E	Tr400x5	10,153-21,756	31.93 in2	20.55 in	15.83 in	2.36 in	0.67 in	100.3 lb
TTHMV-100E	Tr500x5	10,153-21,756	37.72 in2	25.04 in	19.76 in	2.64 in	0.75 in	154.3 lb
TTHMV-142E	Tr710x7	10,153-21,756	47.63 in2	34.25 in	28.03 in	3.07 in	0.98 in	297.6 lb
TTHMV-160E	Tr800x7	10,153-21,756	49.24 in2	37.99 in	31.57 in	3.15 in	0.98 in	354.9 lb